



Cornell University

K. Lisa Yang Center for Conservation Bioacoustics

Unsupervised Machine Learning (exercise)

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Exercise Goals:

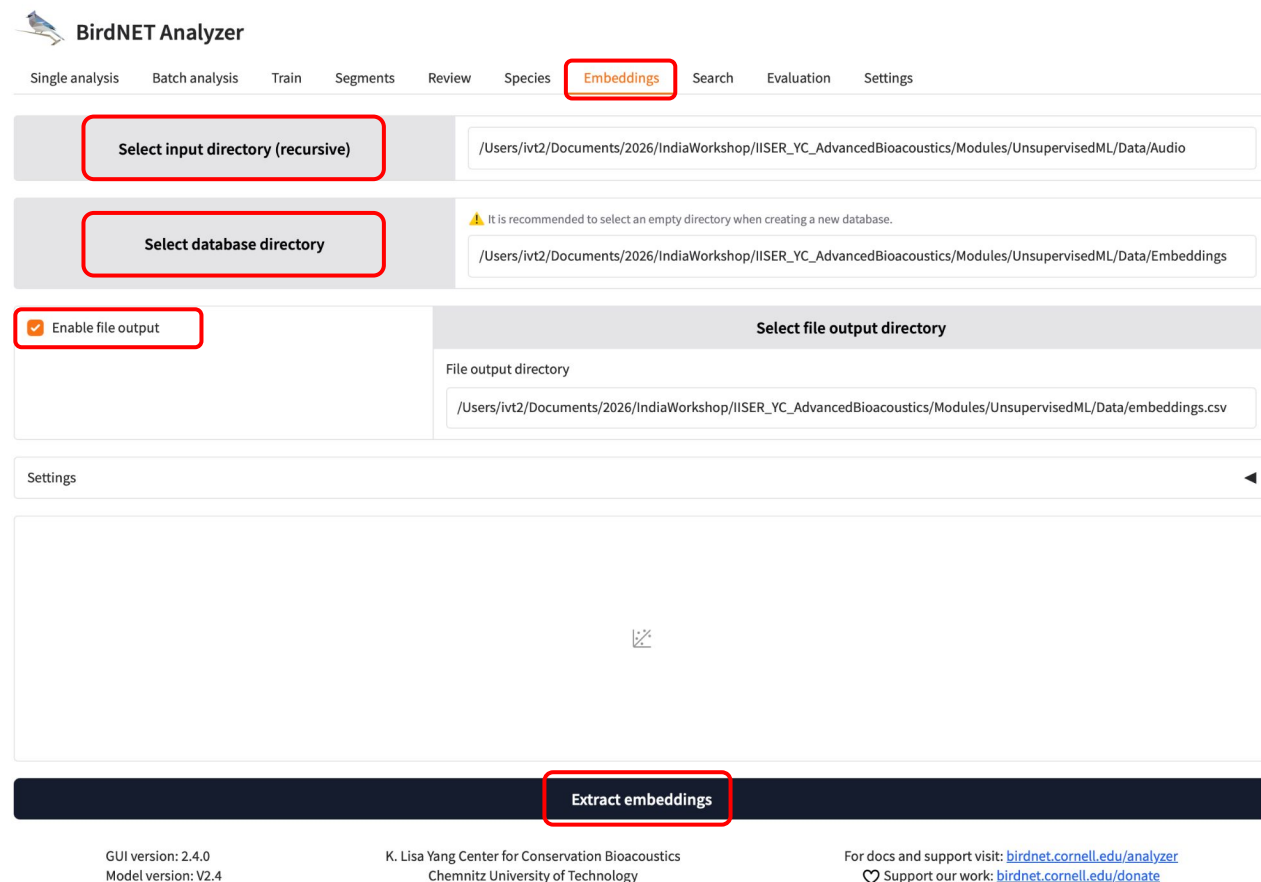
- Calculate **deep-learning-based embeddings** for bioacoustic datasets with BirdNET-Analyzer.
- Visualize these embeddings with **dimensionality-reduction methods** like PCA, t-SNE, and UMAP.
- Examine and interpret **clustering** structure.
- Visualize the **nearest neighbors** of different audio samples.
- Consider **confounding variables**, like background noise.
- Perform **one-shot audio search**.

Exercise Materials:

- **Dataset** containing about orca calls and noise samples.
- **R or Python scripts** for data processing and visualiation.
- **BirdNET-Analyzer**.
- **Tensorflow Visualizer** site link: <https://projector.tensorflow.org/>

In BirdNET Analyzer, use the “Embeddings” tab to calculate embeddings for all audio files.

For “Select database directory”, create and select an empty folder.



The screenshot shows the BirdNET Analyzer web interface. The 'Embeddings' tab is selected in the top navigation bar. The 'Select input directory (recursive)' field is highlighted with a red box. The 'Select database directory' field is also highlighted with a red box, with a warning message below it: 'It is recommended to select an empty directory when creating a new database.' The 'Enable file output' checkbox is checked and highlighted with a red box. The 'Select file output directory' field is highlighted with a red box. The 'Extract embeddings' button at the bottom is highlighted with a red box. The interface includes a settings section at the bottom with a close button.

BirdNET Analyzer

Single analysis Batch analysis Train Segments Review Species **Embeddings** Search Evaluation Settings

Select input directory (recursive) /Users/ivt2/Documents/2026/IndiaWorkshop/IISER_YC_AdvancedBioacoustics/Modules/UnsupervisedML/Data/Audio

Select database directory ⚠ It is recommended to select an empty directory when creating a new database. /Users/ivt2/Documents/2026/IndiaWorkshop/IISER_YC_AdvancedBioacoustics/Modules/UnsupervisedML/Data/Embeddings

☒ Enable file output

Select file output directory

File output directory /Users/ivt2/Documents/2026/IndiaWorkshop/IISER_YC_AdvancedBioacoustics/Modules/UnsupervisedML/Data/embeddings.csv

Settings

Extract embeddings

GUI version: 2.4.0
Model version: V2.4

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For docs and support visit: birdnet.cornell.edu/analyzer
Support our work: birdnet.cornell.edu/donate

Run the provided R or Python scripts to generate **formatted embedding** and **metadata tab-separated files**, or follow along using the files provided.

3.765869140625000000e-02	6.313476562500000000e-01
5.303955078125000000e-02	7.843017578125000000e-02
5.141601562500000000e-01	9.631347656250000000e-02
2.272949218750000000e-01	6.606445312500000000e-01
5.419921875000000000e-01	3.671875000000000000e-01
0.0000000000000000e+00	2.929687500000000000e-01
0.0000000000000000e+00	1.048828125000000000e+00
2.406005859375000000e-01	3.356933593750000000e-01
4.833221435546875000e-03	0.0000000000000000e+00
8.046875000000000000e-01	7.690429687500000000e-01
3.276367187500000000e-01	6.308593750000000000e-01
8.994140625000000000e-01	5.092773437500000000e-01
1.933593750000000000e-01	1.998291015625000000e-01
1.813964843750000000e-01	3.906250000000000000e-02
3.117675781250000000e-01	1.030273437500000000e+00
0.0000000000000000e+00	0.0000000000000000e+00
5.107421875000000000e-01	2.622070312500000000e-01
2.548828125000000000e-01	3.122558593750000000e-01
1.262695312500000000e+00	3.808593750000000000e-01
1.044311523437500000e-01	5.268554687500000000e-01
2.517089843750000000e-01	2.005615234375000000e-01
1.050415039062500000e-01	1.826477050781250000e-02
8.354492187500000000e-01	1.708984375000000000e-02
2.303466796875000000e-01	4.199218750000000000e-01
4.353027343750000000e-01	9.478759765625000000e-02
0.0000000000000000e+00	2.878417968750000000e-01
3.796386718750000000e-01	1.887207031250000000e-01
2.534484863281250000e-02	1.068359375000000000e+00
8.676757812500000000e-01	3.729248046875000000e-02
5.834960937500000000e-02	4.519042968750000000e-01
3.204345703125000000e-02	4.741210937500000000e-01
2.099609375000000000e-02	2.064208984375000000e-01

Category	Day	Channel	Time
noise	20160501	Ch1	10
noise	20160501	Ch1	112
noise	20160501	Ch1	0
noise	20160501	Ch1	101
noise	20160501	Ch1	105
noise	20160501	Ch1	108
noise	20160501	Ch1	115
noise	20160501	Ch1	119
noise	20160501	Ch1	129
noise	20160501	Ch1	133
noise	20160501	Ch1	122
noise	20160501	Ch1	126
noise	20160501	Ch1	14
noise	20160501	Ch1	150
noise	20160501	Ch1	136
noise	20160501	Ch1	140
noise	20160501	Ch1	143
noise	20160501	Ch1	147
noise	20160501	Ch1	168
noise	20160501	Ch1	171
noise	20160501	Ch1	154
noise	20160501	Ch1	157
noise	20160501	Ch1	161
noise	20160501	Ch1	164
noise	20160501	Ch1	17
noise	20160501	Ch1	182
noise	20160501	Ch1	185
noise	20160501	Ch1	189
noise	20160501	Ch1	175
noise	20160501	Ch1	178
noise	20160501	Ch1	192

embedding_values.tsv

TSV Document - 11.8 MB

Information

Created

Modified

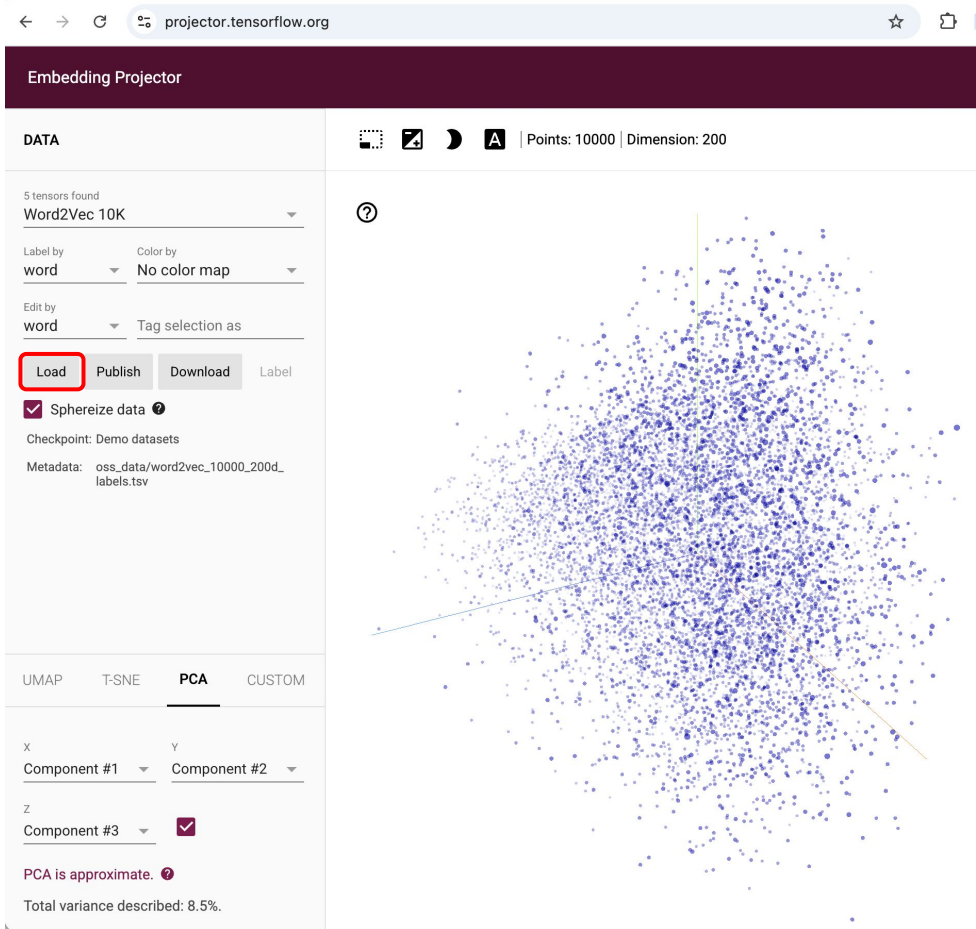
metadata.tsv

TSV Document - 10 KB

Information

Created

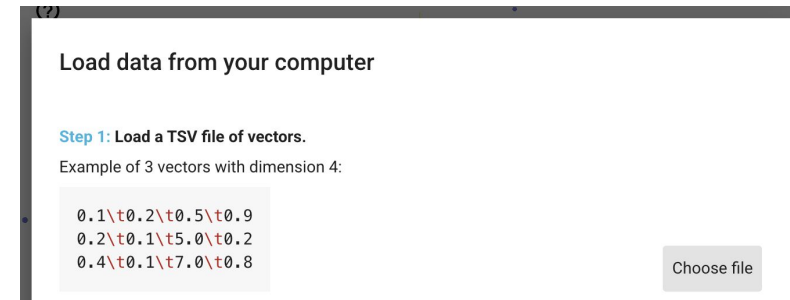
Modified



Open the Tensorflow Projector site:

<https://projector.tensorflow.org/>

Click the “Load” button, and first just enter the embedding file, **without the metadata**.



The projection now represents the audio embeddings of your data, projected to 3D with PCA.

What do you observe? Is there any clustering?

Change the projection method to t-SNE or UMAP.
How does the structure of the point cloud change?

Change the projection method between 2D and 3D.
What is the effect of this change?

UMAP
 T-SNE
 PCA
 CUSTOM

X
Component #1
 Y
Component #2

Z
Component #3
 ☒

PCA is approximate. ?

Total variance described: 40.3%.

Next, use the “load” button to additionally upload the metadata file.

Step 2 (optional): Load a TSV file of metadata.

Example of 3 data points and 2 columns.

Note: If there is more than one column, the first row will be parsed as column labels.

```
Pokémon\tSpecies
Wartortle\tTurtle
Venusaur\tSeed
Charmeleon\tFlame
```

In the left-side panel, select
“Category” as a color option.
You should now see a colored
point cloud.

DATA

5 tensors found

Word2Vec 10K

Label by

Category

Edit by

Category

Load

Publish

☒ Sphereize data ?

Checkpoint: embedding_v

Metadata: metadata.tsv

Color by

No color map

Metadata

Category 3 colors

Day 10 colors

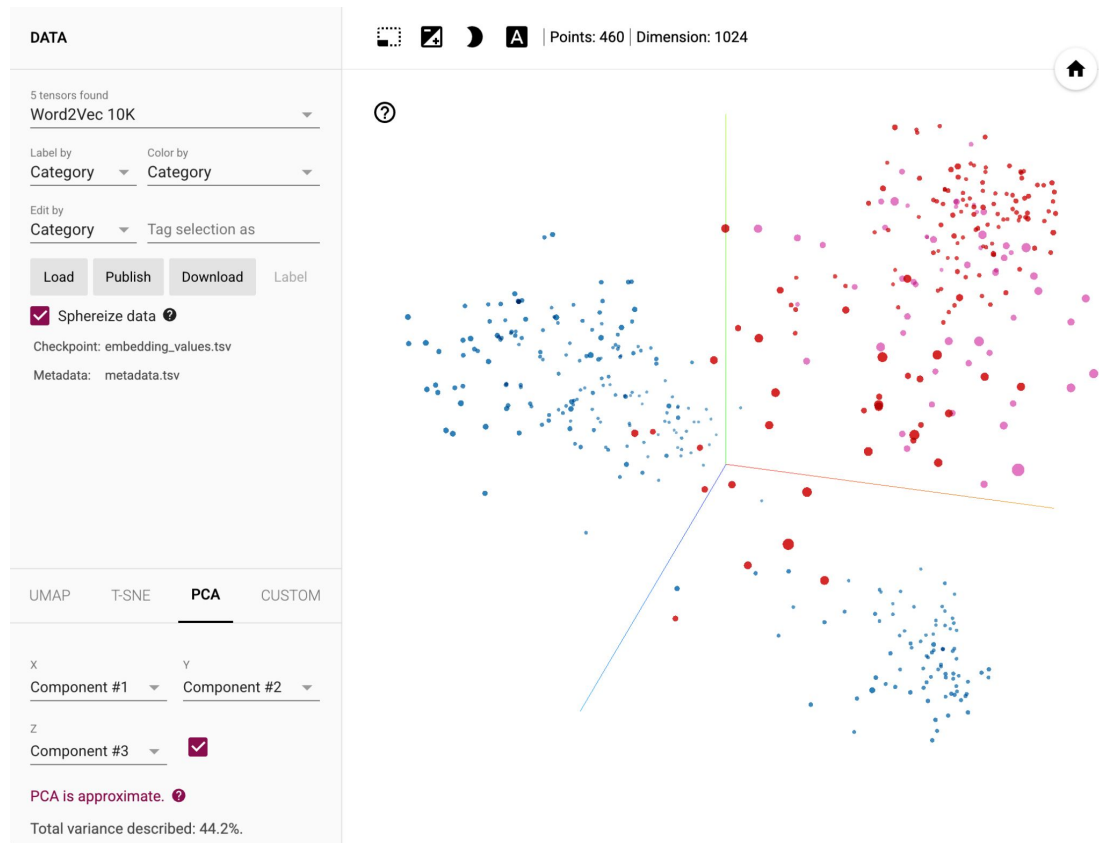
Channel 1 colors

Time gradient

What kind of structure do you observe now?

Does the metadata enable any interpretation of the clusters?

What are several explanations for the clustering we observe?



Next, click on any point in the point cloud. You should see a collection of nearest neighbors appear on the right, and light up in the cloud.

Why aren't all of these "nearest neighbors" always close together in the point cloud?

How could this be useful to bioacoustics?

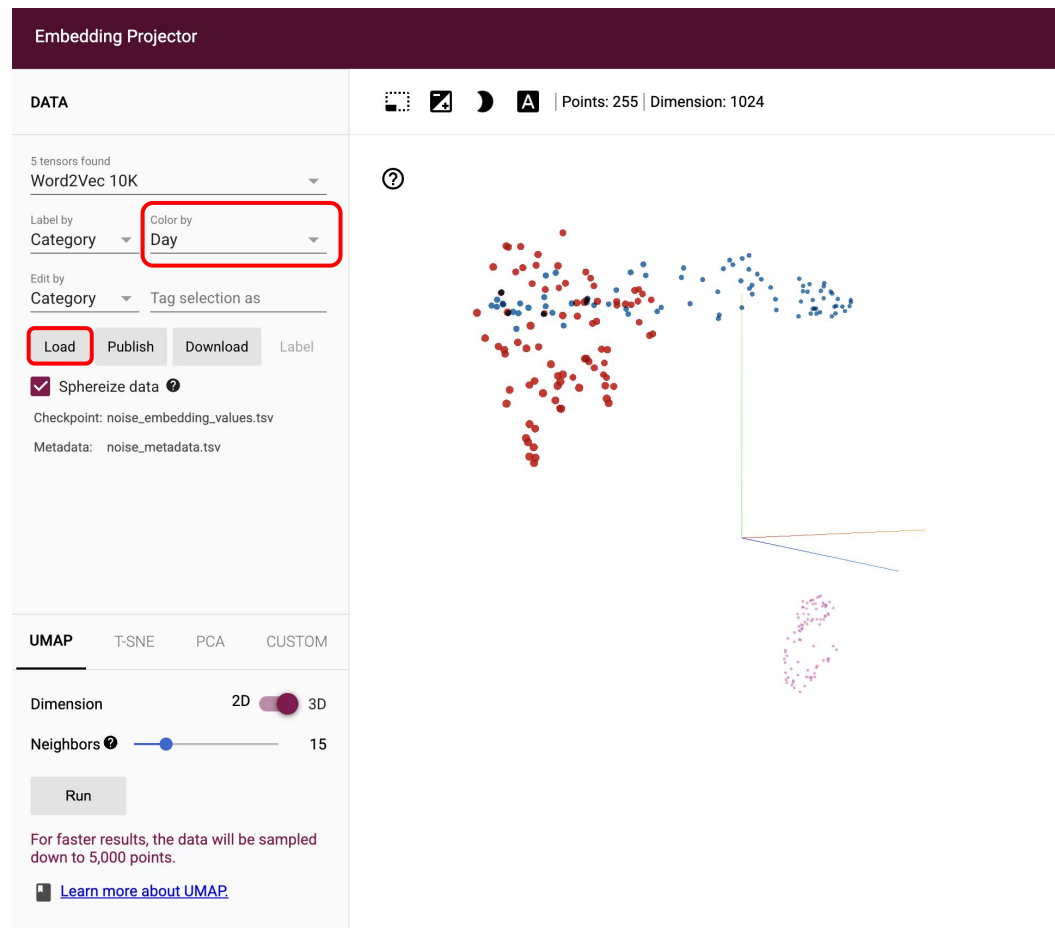


While it can be tempting to interpret clustering as a clear sign of difference in vocalizations, it can also be associated with other (perhaps less interesting) causes, such as background noise patterns.

Revisiting the “load” button, now load the audio embeddings and metadata for **only the noise files**.

Set “Day” as the “Color by” option.

What do you observe?



 **BirdNET Analyzer**

Single analysis Batch analysis Train Segments Review Species Embeddings **Search** Evaluation Settings

Select database

Database
/Users/ivt2/Documents/2026/IndiaWorkshop/IISER_YC_AdvancedBioacoustics/Modules/UnsupervisedML/Data/Embeddings

Embeddings in Database
460

Bandpass frequencies
0 - 15000 Hz

Audio speed factor
1

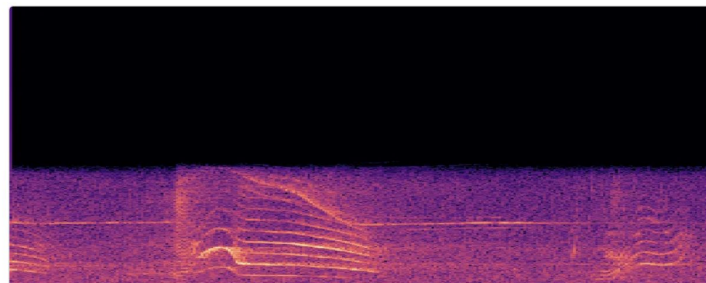
Export selected samples

Select query sample

Query sample

Revisiting BirdNET Analyzer, select the same **“database” folder** we used previously.

Now, select any **“query sample”** that you would like. We will be searching the database for this sample.



Select query sample

Query sample

/Users/ivtz/Documents/zvzb/indiaaworkshop/IISER_YC_AdvancedBioacoustics/modules/UnsupervisedML/Data/Audio/20160512_234836_Ch1_210s_sar.wav

Crop mode

Adjust how to crop the query sample if it is longer than the model input.

☒ center ☐ first ☐ segments

Maximum number of samples to search for

10

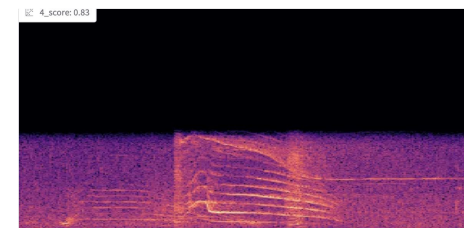
Score function

☒ cosine ☐ dot ☐ euclidean

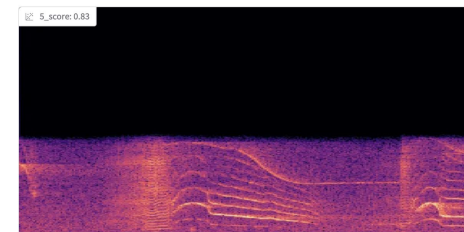
Start search

Click “**start search**”. A collection of similar samples should not appear on the right-hand-side.

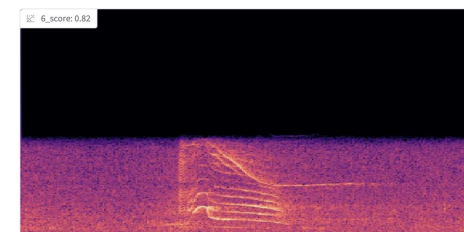
Look through the pages. How do the early samples differ from audio samples in later pages?



Export



Export



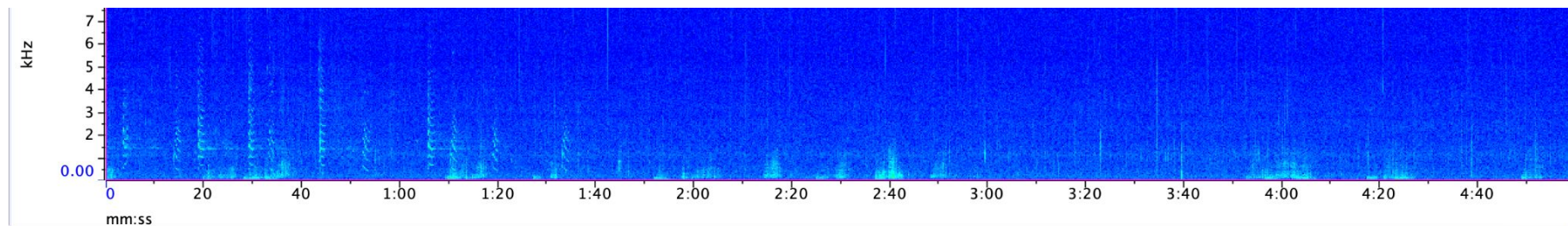
Export

Previous Page

Next Page

Audio search can also be used to search a longer audio recording (effectively, BirdNET Analyzer will slice the recording into shorter time segments).

Take a look at the 5-minute “longer audio” file. We see a few calls in the first two minutes of this file.



Following previous steps, calculate an embedding database for this recording.

(it is helpful to add overlap between windows)

 BirdNET Analyzer

Single analysis Batch analysis Train Segments Review Species **Embeddings** Search Evaluation Settings

Select input directory (recursive)

Select database directory

☐ Enable file output

Settings

Overlap (s)
Adjust the overlap between the samples to extract embeddings from.
0 2.9

Batch size
Number of samples to process at the same time.

Threads
Number of CPU threads to use.

For a workshop project, consider:

Try **calculating embeddings and visualizing clustering structure on your own dataset**. What kinds of results do you find, and are they consistent with what you expect? What is the risk of confounding variables in your data?

Try **running single-shot search for your own dataset**, or for another available bioacoustic recording. Is the search effective? If not, can it be improved by modifying search parameters?

Try **modifying the parameters** in either the Tensorflow Projector or BirdNET Analyzer, such as bandpass filtering, audio speed modification, form of distance measurement, or others. How do these affect your results?